

# BARAA KABBANI

## Computer Science (AI) Undergraduate

Baraaka152@gmail.com | +971 56 722 6128 | Al Majaz 1, Sharjah, UAE | LinkedIn: Baraa Kabbani | GitHub: Baraakabaani

### SUMMARY

CS (AI) undergraduate at the British University in Dubai (2026) with internship experience and strong project work in LLM systems (RAG, GraphRAG, fine-tuning), applied ML, data visualization, and full-stack & mobile development.

### EDUCATION

#### BSc Computer Science – Artificial Intelligence, British University in Dubai (BUID)

2022 – 2026

- Relevant coursework: NLP Applications, Special Topics in AI (RAG, LLM fine-tuning), Pattern Recognition, Machine Learning, Data Visualization, Vision & Robotics
- KU Musabaqat VI (2024) – Khalifa University mathematics competition for top GCC undergraduates

### PROJECTS

#### AI-Based Ultrasonic Attendance System

Capstone Project

- Built a smart attendance system using inaudible ultrasonic chirps (20–21 kHz) to verify student presence in real time — instructor app emits tones, student app listens via a Goertzel-based frequency detector
  - On-device XGBoost classifier (200 trees, F1 = 0.886, AUC = 0.969) trained on 15 sensor & behavioural features to detect spoofing (proxy attendance, abandoned phone); biometric confirmation via Android BiometricPrompt with real-time Firebase Firestore session sync
- Tech: Flutter/Dart, Kotlin, Goertzel DSP, XGBoost, Firebase Firestore, BiometricPrompt

#### GraphRAG Research Assistant – PDF Paper Retrieval Agent

- Built an AI agent combining hybrid retrieval (dense + keyword) and GraphRAG over a 100-paper research corpus — MongoDB (documents), Qdrant (vector search) and Neo4j (knowledge graph) behind a FastAPI backend, fully orchestrated with Docker Compose
- Fine-tuned Llama-3.2-1B with QLoRA on a QA dataset augmented via the Groq API; evaluated output with RAG metrics (faithfulness, answer relevance, context precision)

Tech: Python, FastAPI, Qdrant, Neo4j (Cypher), MongoDB, Docker Compose, Hugging Face, QLoRA/PEFT, Groq API

#### Full-Stack Recruiting Platform with LinkedIn Import Extension

- Developed a comprehensive full-stack recruiting platform enabling clients to create job openings, manage postings, process applications, and showcase candidates through an intuitive interface
- Built a LinkedIn scraping browser extension that automatically imports candidate profile information into the platform, cutting manual data entry for recruiters

Tech: Full-stack JavaScript, REST APIs, browser extension (LinkedIn data import)

#### Bilingual Traffic Sign Recognition (ViT + OCR)

- Fine-tuned ViT-Base on 11,994 real-world UAE/Jordan traffic sign images across 29 classes, achieving 94.82% accuracy (Precision 95.10%, F1 94.90%)
- Integrated EasyOCR for parallel bilingual (Arabic/English) text extraction as a semantic redundancy layer; robustness-tested under fog, low-light, noise & occlusion with Grad-CAM explainability

Tech: Python, PyTorch, ViT, EasyOCR, Grad-CAM, OpenCV

#### RAG System – Respiratory Infection Knowledge Assistant

- Built an end-to-end RAG pipeline answering clinical questions on respiratory infections — document ingestion, chunking, embedding-based semantic retrieval, and LLM generation grounded in retrieved medical context to reduce hallucination

Tech: Python, sentence embeddings, vector database, LLM API

#### Student Group Formation via K-Means Clustering

- Clustered students by skills and GPA for balanced team formation — k = 5 selected via the Elbow Method, StandardScaler preprocessing, PCA visualization, and evaluation with Silhouette, Davies-Bouldin and Calinski-Harabasz metrics

Tech: Python, scikit-learn, K-Means, PCA, pandas

#### Interactive Data Visualization Dashboards – Hotel & Traffic Data

- Built Tableau dashboards and interactive Python visualizations analyzing hotel and traffic datasets, turning raw data into explorable visual insights

### WORK EXPERIENCE

#### Software Development Intern, Lumiver Tech

16-Week Internship

- Contributed to development, debugging and performance optimisation of web applications in a professional team, applying Git workflows across the full software development lifecycle, including client-facing deliverables

#### Freelance Web Developer, Self-Employed

2024 – Present

- Designed and built responsive web apps for clients using React, Node.js and Express.js — gathering requirements, delivering functional solutions and optimising performance

### TECHNICAL SKILLS

- Programming:** Python, JavaScript, Java, C++, Dart, SQL, HTML, CSS
- AI/ML & LLMs:** RAG/GraphRAG, LLM fine-tuning (QLoRA/PEFT), Hugging Face, PyTorch, scikit-learn, XGBoost, ViT, spaCy, EasyOCR
- Data & Visualization:** Tableau, interactive Python dashboards, Qdrant, Neo4j, MongoDB, Firebase Firestore
- Web, Mobile & Tools:** React, Node.js, Express.js, FastAPI, REST APIs, Flutter, Android (Kotlin), Docker Compose, Git/GitHub

### SOFT SKILLS

- Team Collaboration
- Client Communication
- Adaptability & Fast Learning
- Problem-Solving & Debugging
- Time Management
- Bilingual: Arabic & English